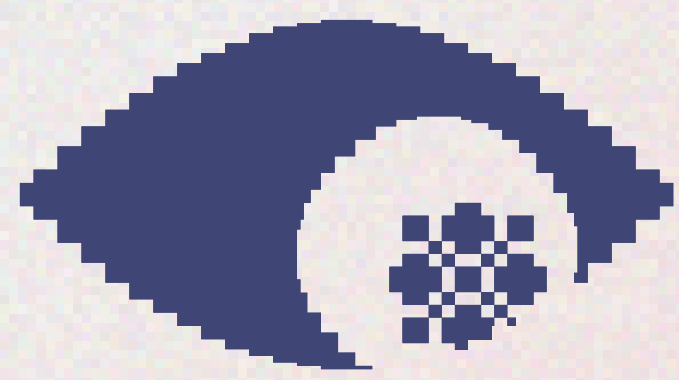




LISBON'S
BEST COURSE
IN SUMMER



HeatAware:

BIOSIGNAL-DRIVEN HEAT STRESS MONITORING WITH AI

AUTHORS

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AFFILIATIONS

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INTRODUCTION

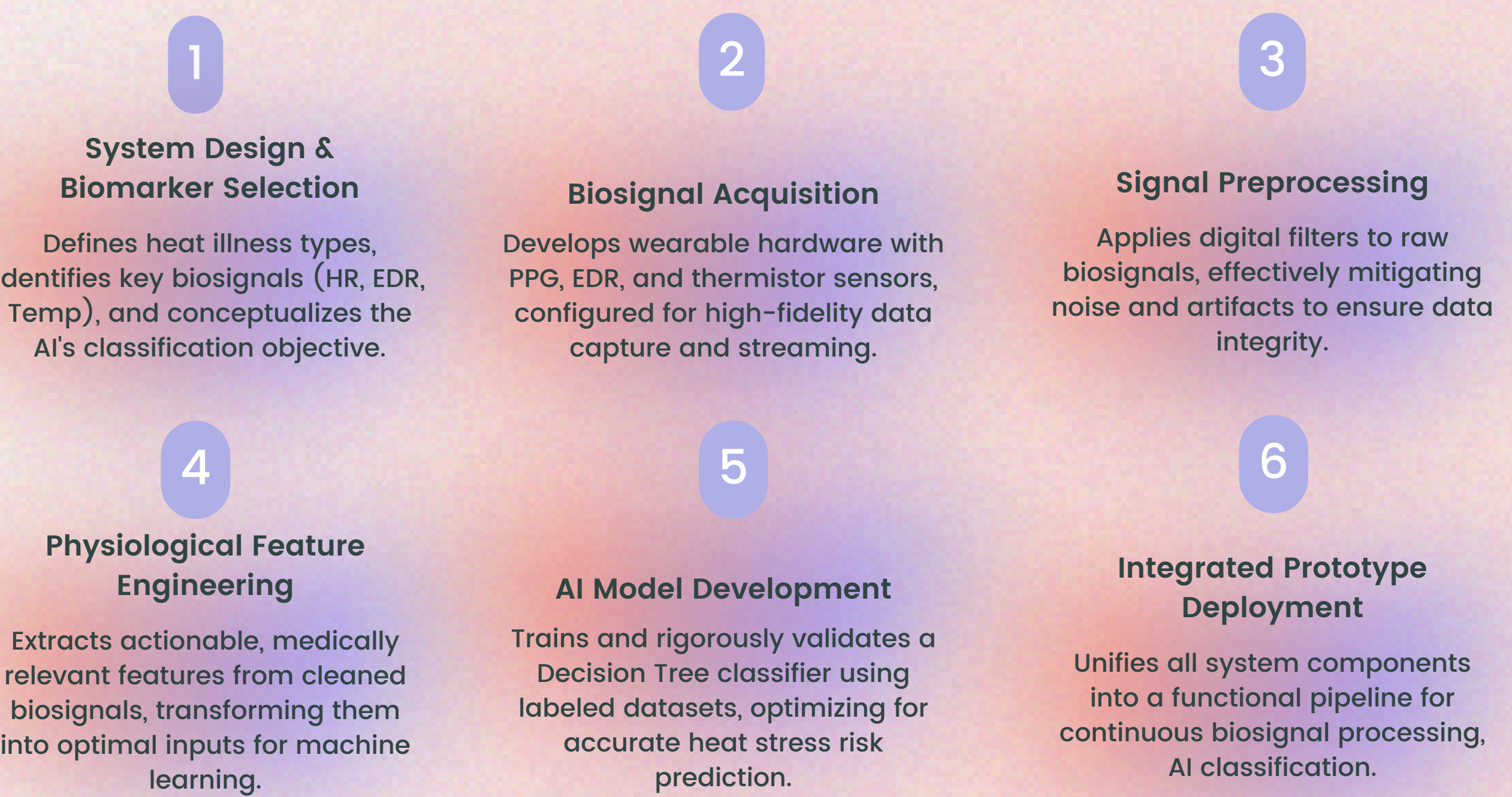
Prolonged exposure to high temperatures can trigger a range of heat-related illnesses, most commonly heat exhaustion and heat stroke. Heat exhaustion is an early-stage condition in which the body begins to struggle with temperature regulation, often marked by symptoms such as heavy sweating, dizziness, weakness, nausea, and a rapid pulse. It serves as a warning sign and, if untreated, can progress into heat stroke: a far more serious, potentially fatal condition.

In heat stroke, the body's core temperature rises above 40°C (104°F), and its natural cooling mechanism shuts down. Sweating ceases, the skin becomes hot and dry, and the person may experience confusion, disorientation, or lose consciousness altogether. Each year, heat-related illnesses lead to over 67,000 emergency department visits and approximately 702 deaths in the United States alone, according to the CDC (2023). Many of these incidents are preventable if early signs are recognized and addressed in time.

OBJECTIVE

To develop a proactive, real-time wearable system that effectively monitors an individual's physiological state to assess heat stress risk. This involves accurately capturing Heart Rate (HR), Electrodermal Activity (EDA), and both body & ambient temperatures; applying advanced digital signal processing for noise reduction; extracting a comprehensive set of relevant physiological features; and utilizing an interpretable Artificial Intelligence model, specifically a Decision Tree classifier, to provide timely detection of impending heat exhaustion and the onset of heat stroke.

METHODOLOGY

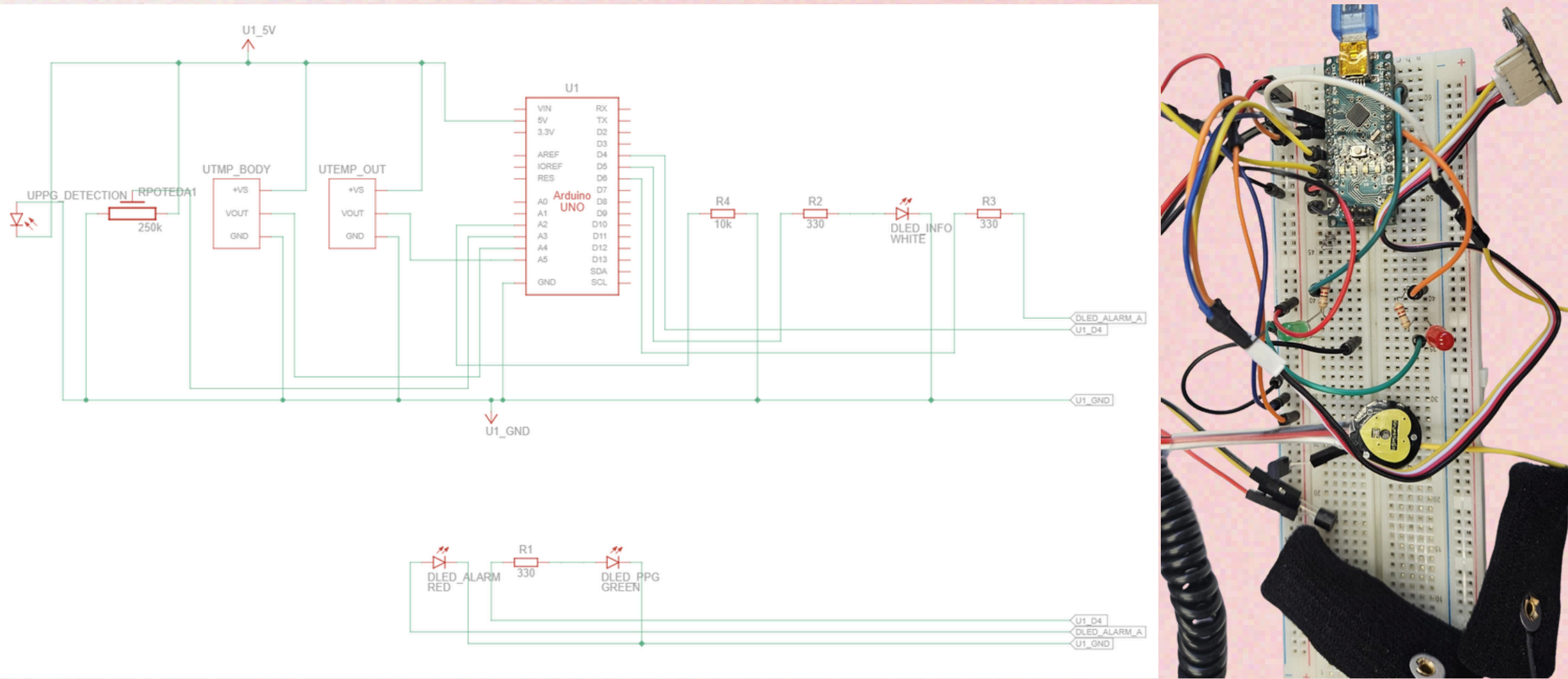


DESIGN

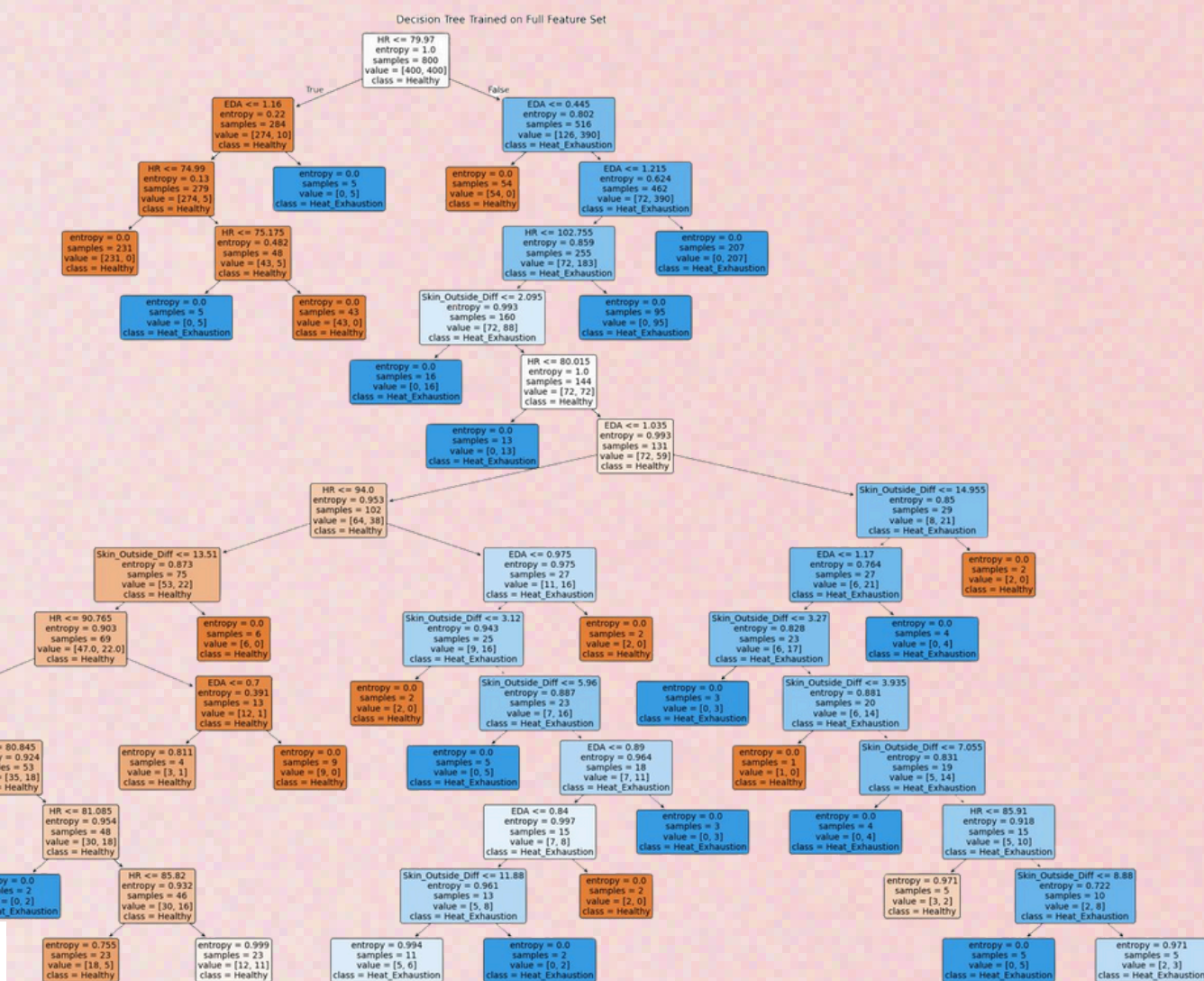
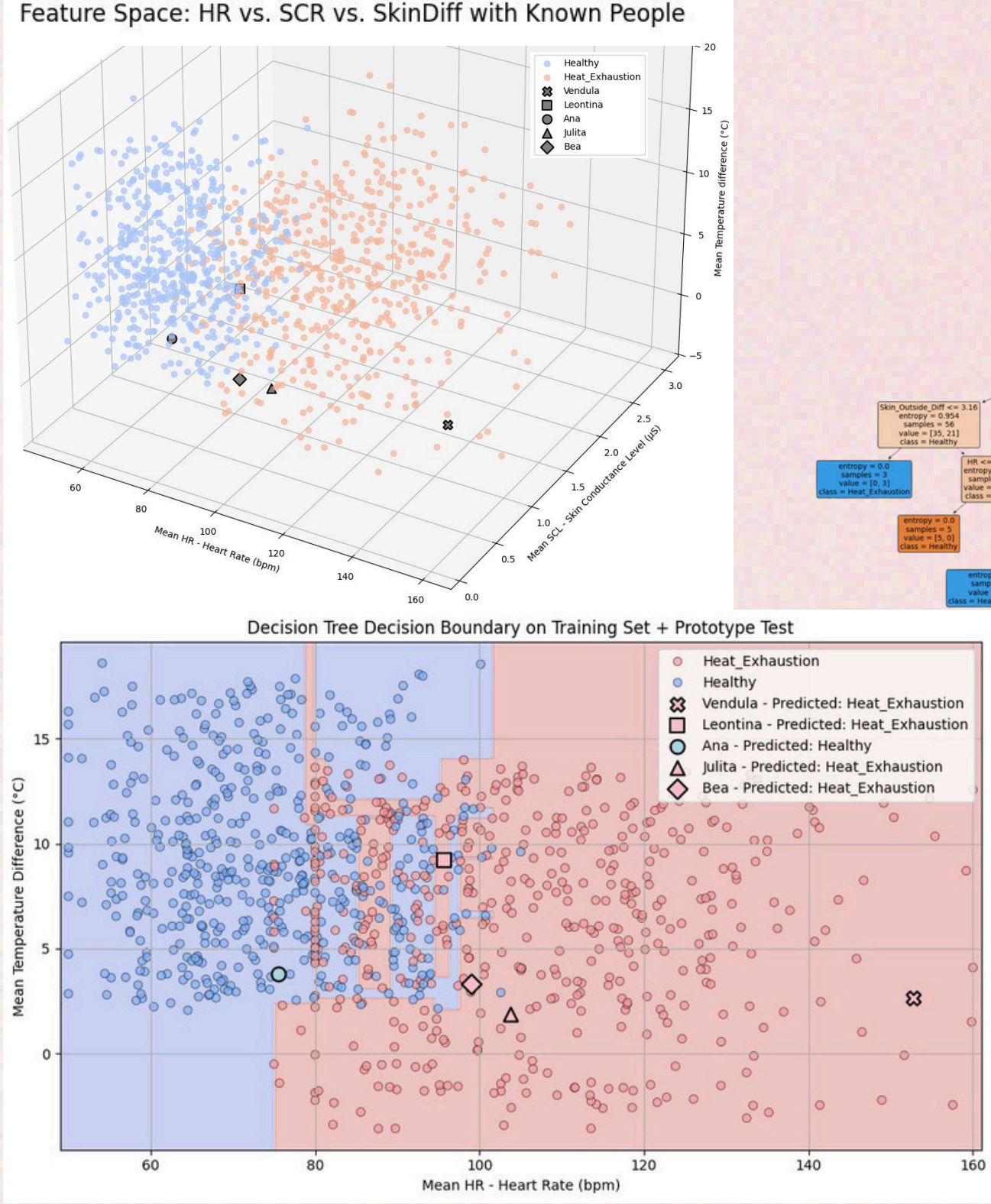


ANALYSIS

Breadboard Prototype



Trained Classifier AI Model

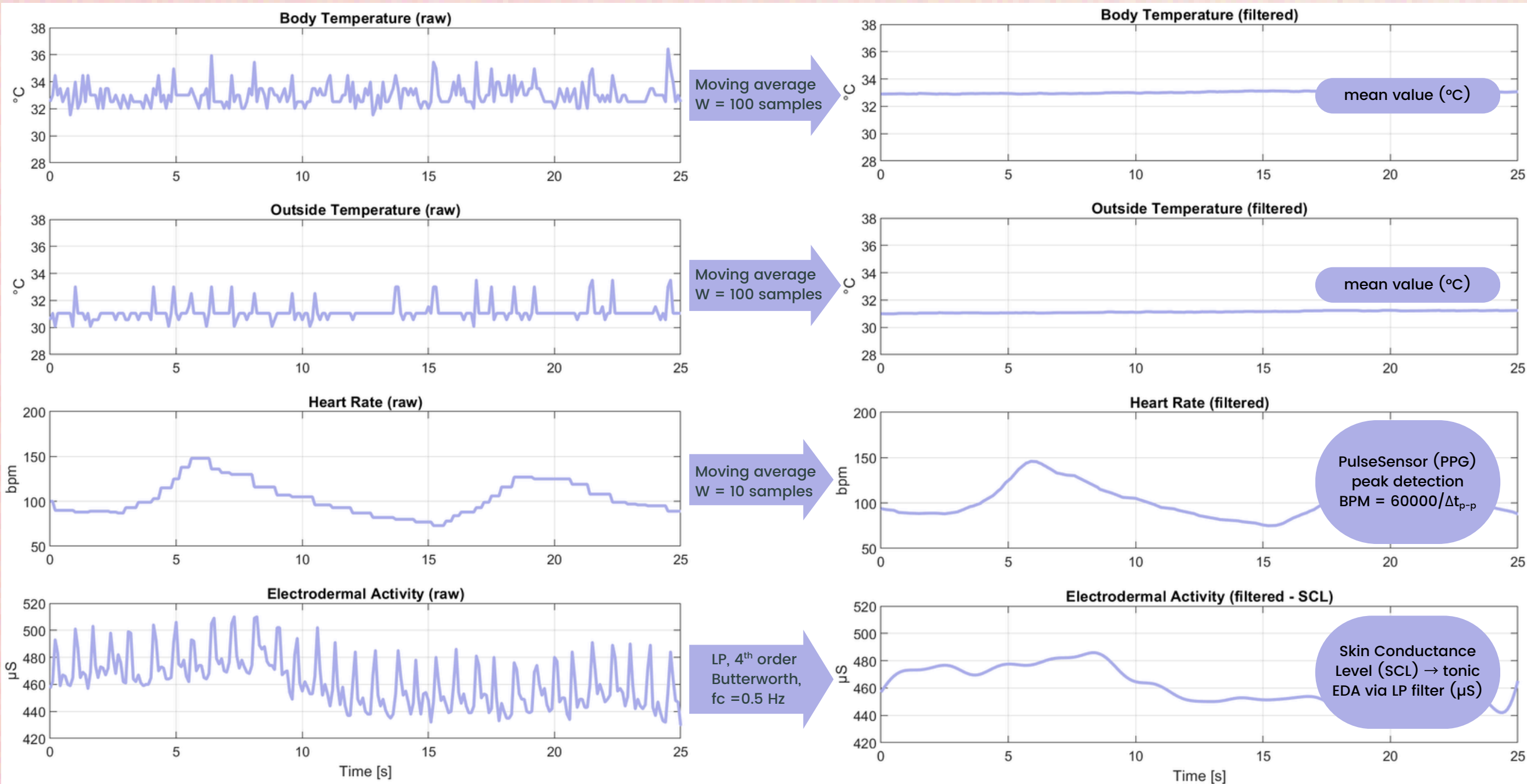


DECISION TREE

- Features used: Mean HR, Mean SCL, Mean TempDiff
- Best DT parameters: 'criterion': 'entropy', 'splitter': 'best', 'max_depth': 14, 'min_samples_split': 10
- Metrics: F1: 0.97, Accuracy: 0.95, AUC: 0.96
- Validation Accuracy with noise:
 - No noise:
 - STD 0.5: 80.0%
 - STD 1.5: 64.50%
 - STD 3.0: 56.0%

From Raw Biosignal to Filtered Signal and Feature Set

- Raw data: 10 Hz sampling, 20-second measurement



CONCLUSION

In this project, we developed a wearable prototype designed to detect the risk of heat stroke by monitoring electrodermal activity (EDA), heart rate (HR), outside and body temperature. The system filters raw sensor signals, extracts key physiological features, and applies a Decision Tree (DT) classifier to determine whether the individual is currently experiencing heat stroke risk. The final model using mean heart rate, mean skin conductance level, and temperature difference achieved high accuracy (95%) and F1-score (0.97), but its performance decreases under noisy input conditions, indicating sensitivity to sensor signal quality.

This work demonstrates the potential of wearable sensing combined with machine learning for real-time health monitoring and early prevention.